Exp no: 09 Internetworking Implementation
28.9.26 Implementation of Subnetting in CISCO PACKET TRACKER Simulator

AIM:-

To implement of Subnetting in CISCO PACKET TRACKER Simulator.

Procedure:-

Classless IP Subnetting is a technique that allows for more efficient use of IP addresses by allowing for subnet masks that are not just the default.

Creating a network Topology:-

To first step in implementation classless IP Subnetting is to create a network topology in packet Tracer. To create a network topology "new" button.

Adding the devices:-

Once we have created our network topology we can add devices to it. Here, we will be adding routers. Then connect the devices by dragging a cable from one device.

#enable
configure terminal
interface terminal fast Ethernet 0/0
ip address
no shutdown
exit.

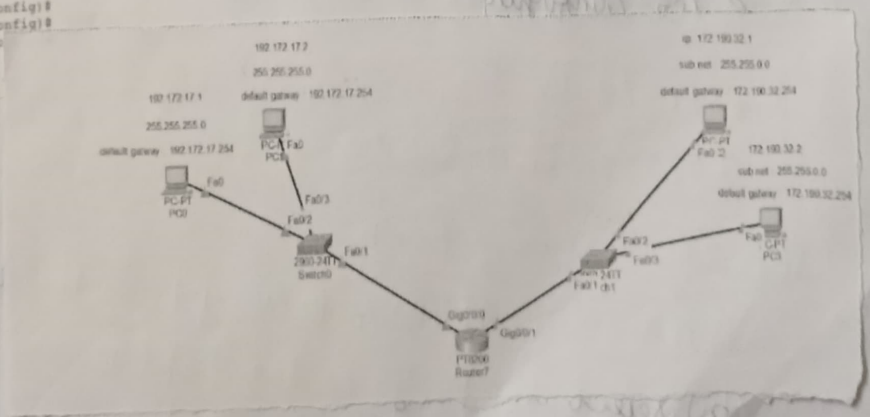
Subnetting:-

To Subnet the network address of 192.168.1.0/24 to provide enough space for atleast 5 address for end devices.

Testing the network:-

Now that our network topology is configured we can test the network. Open a command prompt on each PC and try to ping the other PC. We can also use the "Ping" command to test connectivity.

```
Router>
*Sep 24, 11:17:27.1717: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0/1, changed state to up
Router>
Router>
Router>
Router>
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.172.17.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 172.190.32.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(co
```



Result:-

Thus the implementation of Subnetting in CISCO packet Tracker Simulator.